Presentation 5

• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallOutcomesFinal_9Dec_EDA2.pdf

Presentation Video:
** Video was too large to push to GitHub – emailed to Krish instead and posted .txt with link
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Dec9_EDASimple2.txt

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/CallOutcomeTagging_9Dec_Vivienne.ipynb (I did not directly code for this assignment, but I did help write out some of the dictionaries for tagging for Vivienne to run on her machine)

• What did you do?

I wrote half of the dictionary tagging for the outcome types and subtypes for Vivienne to compile together in her continuation of Krish's outcome tagging code. I also cross-checked the totals and proportions for the Call Breakdown by Call Type and Call Outcomes by Call Type reports.

• How does it help the project?

This provided a top-down view of how call volumes and outcomes are distributed across categories. It also streamlined the workflow as Vivienne and Sabrina built the Python functions to assign outcomes and subtypes to all contact session IDs.

• Issues faced (if any)

We faced some issues trying to figure out how to transfer our previous work in R to Krish's code in Python.

• Attempts to resolve issues (if any)

We decided it would be easiest to just keep it all consistent in dictionary format in Python to easily map outcomes to contact session IDs.

• Issues resolved (if any)
N/A

• Next steps
N/A (end of quarter)

• References (Mention if you built up on someone else's work)

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/Caller_type_breakdown_1Dec_Krish.pdf

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/caller_type_tag_1Dec_Krish.ipynb

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Clarifications/CAR_data_Activity_name_tagging_1Dec_for_final_ppt.xlsx

Presentation 4

• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomeTagging_Nov18_EDASimple2.pdf

Presentation Video:
** Video was too large to push to GitHub – emailed to Krish instead and posted .txt with link
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov18_EDASimple2.txt

Code:
http://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeTagging_Nov17_CarolineCorr.R (worked together)

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeVisualization_Nov18_HaileeKim.R

• What did you do?

Nikole, Caroline, and I split up the ~90 last activities in the Excel sheet and wrote the case when statements to map all of their outcomes/sub-types.

Then I visualized the percentages of positive, negative, and neutral outcomes and their split into the above sub-categories for calls ending in the following menus: I) Main menu, II) Pre-legal Seniors menu, III) Legal menu, IV) Family menu, V) Housing menu, VI) Benefits menu, VII) Employment menu, VIII) Immigration menu, IX) Other legal menu, X) HIV menu. I first standardized the dataset by recoding each IVR menu into one of ten meaningful menu categories and enforcing ordered factor levels for the three outcome types (Positive, Neutral, Negative). I then aggregated the data to compute the distribution of outcomes within each menu, producing a summary table that reports both counts and percentages. This enabled the creation of an initial faceted donut-chart visualization showing overall outcome proportions by menu. To explore results more granularly, I performed a second aggregation that breaks each outcome into its underlying sub-types and calculated within-outcome percentages. For each menu category, I generated a row of three donut charts — one for Positive, Neutral, and Negative outcomes — using distinct color palettes to clearly differentiate the sub-categories within each outcome. The plotting was automated through a loop that filters the data for each menu, constructs the corresponding donuts with consistent formatting, combines them into a horizontal layout, and optionally saves each finalized figure to the graphics directory.

• How does it help the project?
My work enabled our team to present our tagging results in a coherent framework. Our team's analysis provides LegalAid with actionable insight into how well the existing routing system meets client needs. These findings help the organization assess whether the current menus, queue staffing, and messaging structure are achieving their intended purpose and highlight where refinements could reduce call failures and increase the proportion of callers who successfully reach assistants.

• Issues faced (if any)
N/A

• Attempts to resolve issues (if any)
N/A

• Issues resolved (if any)
N/A

• Next steps

To move this analysis forward, we should first resolve the remaining unclear or untagged rows in the activity file. Discussing these with Cynthia, Mark, or the EDA 1 team may help reduce the volume of unknown outcomes and sub-outcomes. We also identified low positive outcomes for several menus, and part of this may be due to how outcomes are currently attributed. Right now, each call is assigned to the last menu the caller accessed, but many calls that ultimately become positive actually flow into Queue, Courtesy Callback, or other options. As a result, some successful calls may be getting misclassified or lost when they transition out of the original 10 menu categories. A next step is to determine how we should treat these additional call flows so we accurately capture the full path to positive outcomes rather than undercounting them. Clarifying this will allow us to better represent caller journeys, interpret menu effectiveness, and refine the outcome tagging structure going forward.

• References (Mention if you built up on someone else's work)

http://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeTagging_Nov17_CarolineCorr.R (my visualization script must be run after this file)

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Clarifications/CAR_data_Activity_name_tagging_10Nov.xlsx

For data import:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

Presentation 3

• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallMenuOutcomes_4Nov_EDATeam2.pdf

Presentation Video:
** Video was too large to push to GitHub – emailed to Krish instead and posted .txt with link
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov5_EDASimple2.txt

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/ImmigrationMenuOutcomes_Nov4_HaileeKim.R

Document for Outcomes & Rules:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomesRulesDoc_Nov5_EDASimple2.pdf

• What did you do?

In this analysis, I filtered the CAR dataset to immigration-related calls occurring after March 15, 2025, defined as any contact session entering the LegalImmigrationMenu flow. I then cleaned and standardized termination reasons, first trimming whitespace inconsistencies and collapsing duplicate labels such as "NA", "N/A", and missing values into a single “NA” category, and consolidating variants of system-timeouts like “RONA Timer Expired” and “RONA_TIMER_EXPIRED” into one unified label. Calls were then divided into two routing paths: **Case 1** — those that reached the Trafficking Voicemail Transfer queue, and **Case 2** — all other immigration routing outcomes under Immigration/Immigration SP queue (e.g., Visa, Crime Victim, Asylum, Court, Naturalization). To accurately represent the final outcome of each call, I grouped by contact_session_id and retained only the latest activity timestamp, ensuring we captured how the call truly ended. Outcomes were classified using team-approved criteria (documented in the Outcomes & Rules reference): **Positive** when callers reached an agent and the call ended normally (e.g., “Agent Left” or “Customer Left” while connected to an agent); **Negative** when callers failed to reach an agent because of abandonment or a system/agent-side failure (e.g., “Customer Left” with no agent connection, “AGENT_UNAVAILABLE,” “NO_ANSWER_FROM_AGENT”); and **Neutral** when callers received recorded guidance, were out-of-scope, or unreachable. I then summarized the distribution of these outcomes across Case 1 and Case 2 and created visualizations — stacked percentage bars, side-by-side counts, and an overall donut chart — to evaluate whether the Immigration Menu routing is successfully helping callers connect with legal assistance.

• How does it help the project?

This work helps the broader project by identifying how effectively the Immigration Menu routes callers toward meaningful assistance. By isolating call paths and classifying outcomes into Positive / Neutral / Negative, we can clearly see whether callers are reaching an agent or dropping off due to abandonment or system bottlenecks. The analysis highlights where callers are getting stuck — especially in the “Other Immigration Routes” branch, which shows a high share of negative outcomes — and pinpoints operational issues like unavailable agents or transfers to voicemail with no follow-up. These insights help Legal Aid Chicago evaluate whether the current routing process is achieving its intended purpose and where improvements to menu options, queue staffing, or message design could reduce call failures and increase successful connections.

• Issues faced (if any)
N/A

• Attempts to resolve issues (if any)
N/A

• Issues resolved (if any)
N/A

• Next steps

Because the *Other Immigration Routes* category (Case 2) represents the largest call volume and the highest proportion of negative outcomes, I would recommend evaluating whether the grouped immigration sub-options (Visa, Crime Victim, Asylum, Immigration Court, Naturalization) should be separated into individual queues. This would help the team pinpoint which topics are driving the most call drop-offs, identify where advocate shortages or delays are occurring, and improve routing accuracy and ultimately agent connection rates.

If the root cause is true agent unavailability, these insights could support staffing or process changes. However, if agents are not actually missing and instead agent data is not being captured properly in CAR, this would provide justification for addressing data completeness issues with the vendor.

I will also cross-reference these results with summary reports generated from the All Calls dataset to verify consistency and flag any differences as potential data integrity concerns.

• References (Mention if you built up on someone else’s work)

For data import:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

Presentation 2

• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/CallDurationComparison_22Oct_HaileeKim.R

• What did you do?
I coded and created a Median Call Duration by Hour graph to visualize and validate the consistency of inbound call durations between the CAR and All Calls datasets. This process involved cleaning and filtering both datasets to include only records after March 15, 2025, standardizing timestamps to the America/Chicago timezone, and calculating duration metrics for each dataset. I summarized the median call duration per hour, combined the two sources, and plotted them together for comparison. In addition to the coding work, I also prepared presentation slides for Objective 1.2, clearly illustrating the overlap between datasets and explaining the rationale behind key data processing choices.

• How does it help the project?
This work directly supports Objective 1, which aims to ensure consistency and reliability in inbound call duration reporting across multiple data sources. By comparing CAR and All Calls datasets after March 15, 2025, I confirmed that both show aligned hourly patterns, suggesting strong data integrity and synchronization between systems. The analysis revealed that median call durations peak around 8 AM - corresponding to the start of operational hours - and gradually decline toward the end of the day, which reflects expected operational trends. Additionally, I verified that differences between answer_time and start_time in the All Calls dataset have minimal impact, simplifying future analysis of call timing variables. Finally, I identified that refining the hour variable logic in the CAR dataset - using start hour before 17:00 and end hour afterward - improves classification accuracy for extended evening calls.

• Issues faced (if any)
N/A

• Attempts to resolve issues (if any)
N/A

• Issues resolved (if any)
N/A

• Next steps

Durations during operational hours showed strong alignment between the two datasets, while some discrepancies appeared outside of these hours. To improve overall accuracy, the next step is to further investigate calls occurring during non-operational periods to understand how they are being captured, timestamped, and classified. This deeper review will help identify whether these differences stem from system logging behaviors, after-hours routing processes, or data extraction inconsistencies.

• References (Mention if you built up on someone else's work)

For data import:

CAR:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

All Calls:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/allcallsimport_Oct7_LoganRoever.R

Presentation 1

• Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/AllInboundCalls_6Oct_CarolineCorr.R

• What did you do?

I pulled relevant code from the previous quarter and developed strategies to tackle Objective 1 (comparing and visualizing the number of calls between the CAR dataset and the corresponding inbound calls in the All Calls dataset). Then I worked with team members to import the files correctly and code the first plot. I also designed the team presentation and compiled the emerging questions that highlighted the issues we faced and informed potential next steps. After the Friday check-in meeting, I added additional comments on handling daylight savings using the with_tz function, and on creating a month_year column to facilitate joining and comparing tables.

• How does it help the project?

It led to a solid visualization of Objective 1 and helped identify potential reasons for the remaining mismatches between the All Calls and CAR datasets. Also, it contributed to making our work more reproducible.

• Issues faced (if any)

Our team initially struggled to correctly read the data into R. We divided into two groups based on our preferred programming languages - I was on the R team with Caroline and Nikole. We followed the approach we had learned in our sequence class, where all datasets are stored in the same local folder and imported as needed. For this reason, we pushed the datasets to GitHub and worked from the /data repository.

After our Friday check-in meeting, we learned that it was possible to incorporate the OneDrive location in the Jupyter notebooks. However, we encountered the error message “The specified directory does not exist,” and we are still troubleshooting this recommended but unfamiliar setup. Because the combined dataset was too large to include in the repository and we wanted to avoid the confusion of multiple people editing the same R scripts simultaneously - a problem we had faced in our sequence class - we decided to code collaboratively on one laptop (Caroline's). All of us met in person at Main and worked together throughout the process.

• Attempts to resolve issues (if any)

I pulled relevant code from the previous quarter of how to import the datasets in R and we also consulted ChatGPT to help edit the code to include CSVs. We also decided to download the All Calls data locally from Vivienne's version in a shared Google drive instead of starting that R import code from scratch.

• Issues resolved (if any)

We were able to import the data but we were told from Krish that we must not push your datasets to GitHub, so fundamentally we're still figuring out how to correctly import the datasets.

• Next steps

After the pre-grading feedback, we've filtered 6 relevant numbers to CAR data. To address the remaining mismatch in inbound call numbers, we plan to further explore whether understanding the category names and filtering by "User Type" could improve accuracy. We can also take a closer look at the timestamp discrepancies between "Activity Start Timestamp" and "Start Time/Answer Time."

• References (Mention if you built up on someone else's work)

We used the R data import code from spring quarter to get started loading in the CAR data.

Link:
https://github.com/arvindkrishna87/STAT390_SP25_LegalAid/blob/main/Presentations/explore.R